

APO-ALLOPURINOL

Allopurinol Tablets USP 100mg and 300mg

Xanthine Oxidase Inhibitor

Pharmacology: Allopurinol is structural analogue of the natural purine base, hypoxanthine. It is a potent inhibitor of xanthine oxidase, the enzyme responsible for the conversion of hypoxanthine to xanthine and of xanthine to uric acid. When taken orally, allopurinol is rapidly absorbed and rapidly metabolized. The main metabolite is oxypurinol, which is itself a xanthine oxidase inhibitor. Allopurinol and its metabolites are excreted by the kidney but the renal handling is such that allopurinol has a plasma half-life about 1 hour, whereas that of oxypurinol exceeds 18 hours. Thus, the therapeutic effect can be achieved by a once a day allopurinol dosage in patients taking 300mg or less/day. Administration of allopurinol generally results in a fall in both serum and urinary uric acid within 2 to 3 days. The magnitude of this decrease can be manipulated almost ad lib since it is dose dependent to a limited extent. A week or more of treatment with allopurinol may be required for full effects of the drug to be manifest since the serum uric acid concentration falls gradually; likewise uric acid may return to pretreatment concentration slowly, usually after a period of 7 to 10 days following cessation of therapy. This reflects primarily the slow accumulation and clearance of oxypurinol. In some patients, particularly those with tophaceous gout, a significant fall in urinary uric acid excretion may not occur. It has been postulated that this fall may be due to the mobilization of urate from the tissue deposits as the serum uric acid concentration begins to fall. It has been shown that reutilization of both hypoxanthine and xanthine for nucleotide and nucleic acid synthesis is markedly enhanced when their oxidations are inhibited by allopurinol. This reutilization and the normal feedback inhibition which would result from an increase in available purine nucleotides serve to regulate purine biosynthesis, and, in essence, the defect of the over-producer of uric acid is thereby compensated. Accompanying the decrease in uric acid produced by allopurinol is an increase in serum and urine concentrations of hypoxanthine and xanthine. Plasma concentrations of these oxypurines do not, however, rise commensurately with the fall in serum urate concentrations and are often 20 to 30% less than would be expected in view of urate concentrations prior to allopurinol therapy. This discrepancy occurs because renal clearance of the oxypurines is at least 10 times greater than that of uric acid. In addition, normal urinary purine output is almost exclusively uric acid, but after treatment with allopurinol, it is composed of uric acid, xanthine, and hypoxanthine, each having independent solubility. Thus the risk of crystalluria is reduced. Alkalinization of the urine increases the solubility of the purines, further minimizing the risk of crystalluria. The glomerular filtration rate and urate clearance in patients receiving allopurinol do not differ significantly from those obtained prior to therapy. The rapid renal clearance of oxypurines suggests that allopurinol therapy should be of value in allowing a patient with gout to increase his total purine excretion. Innate deficiency of xanthine oxidase, which occurs in patients with xanthinuria as an inborn error of metabolism, has been shown to be compatible with comparative well-being. While urinary concentrations of oxypurines attained with full doses of allopurinol may in exceptional cases equal those (250 to 600mg/day) which in xanthinuric subjects have caused formation of urinary calculi, they usually fall in the range of 50 to 200mg and no evidence of renal damage has been observed clinically. Xanthine crystalluria has been reported in a few exceptional cases. Reproductive studies in rats and rabbits indicated that allopurinol did not affect litter size, the mean weight of the progeny at birth or at 3 weeks postpartum, nor did it cause an increase in animals born dead or with malformations.

Indications: The treatment of gout, either primary, or secondary to hyperuricemia which occurs in polycythemia vera, myeloid metaplasia or other blood dyscrasias. Also indicated in the treatment of primary or secondary uric acid nephropathy with or without accompanying symptoms of gout. May be useful in patients with gouty nephropathy, in those who form renal urate stones, and in those with unusually severe disease. May be given prophylactically to prevent tissue urate deposition or renal calculi as well as acute urate nephropathy and resultant renal failure in patients with leukemias, lymphomas or other malignancies who are receiving radiation therapy or antineoplastic drugs with their resultant elevating effect upon serum uric acid concentrations. To prevent occurrence and recurrence of uric acid stones and renal calcium lithiasis in patients with hyperuricemia and/or hyperuricosuria.

Route of administration: Oral

Contraindications: Allopurinol should not be given to patients who are hypersensitive to it or who have previously developed a severe reaction to this drug. Should not be given to nursing mothers and children (except in those with hyperuricemia secondary to malignancy or genetic disorders of purine metabolism).

Warning: Allopurinol should be discontinued at the first appearance of skin rash or other signs which may indicate an allergic reaction. Hypersensitivity to allopurinol usually appears after some weeks of therapy, and

more rarely immediately after beginning treatment. In some instances, a skin rash may be followed by more severe reactions such as exfoliative urticarial and purpuric lesions as well as Stevens-Johnson syndrome, and/or generalized vasculitis, irreversible hepatotoxicity and even death. Hypersensitivity reactions, frequently marked by fever and eosinophilia, usually begin 2 to 4 weeks after start of therapy and appear related to pre-existing renal dysfunction, elevated oxypurinol plasma levels and/or concurrent thiazide therapy. Periodic liver function tests, renal function tests and complete blood cell counts should be performed in all patients on allopurinol. Alterations in liver function test results, including transient elevation of serum alkaline phosphatase, urinary urobilinogen, serum elevations of serum alkaline phosphatase, urinary urobilinogen, serum AST (SGOT) and ALT (SGPT) and decreases in sulfobromophthalein excretion have occurred in some patients. Reversible hepatomegaly, hepatocellular damage (including necrosis), granulomatous changes, hepatitis and jaundice have also occurred. Observe patients with impaired renal or hepatic functions carefully during the early stages of allopurinol administration and withdraw the drug if increased abnormalities in hepatic or renal functions appear. In patients receiving mercaptopurine or azathioprine, the concomitant administration of 300 to 600 mg/day of allopurinol will require a reduction in dose to approximately 33% to 25% of the usual dose of mercaptopurine or azathioprine since their elimination will be prolonged. Make subsequent adjustment of doses of mercaptopurine or azathioprine on the basis of therapeutic response and any toxic effects.

Occupational hazards: Drowsiness may occur. Patients should be cautioned not to engage in activities where alertness is mandatory until their response to the drug is known.

Children: Allopurinol should not be given to children except those with hyperuricemia secondary to malignancy or with Lesch-Nyhan syndrome, because safety and effectiveness have not been established in other conditions.

Pregnancy: Allopurinol is not recommended for use during pregnancy or in women of childbearing potential unless the potential benefits outweigh the possible risks to the fetus.

Lactation: See contraindications.

Precautions: Because therapy with allopurinol is not without some serious risks, the drug is used for the management of gout when uricosurics cannot be used because of adverse effects, allergy, renal insufficiency or inadequate response, when there are visible tophi or radiographic evidence of uric acid deposits and stones; when urinary uric acid excretion is greater than 510 to 540 $\mu\text{mol/L}$ (8.5 to 9mg/100ml) and a family history of tophi and low urate excretion exists. Acute gouty attacks may be precipitated at the start of treatment with allopurinol in new patients and these may continue even after serum uric acid concentrations begin to fall. Prophylactic administration of colchicine (0.6mg twice daily) or a nonsteroidal anti-inflammatory drug is advisable, particularly in new patients and in those where the previous attack rate has been high. In addition, the patient should start with a low dose of allopurinol (100 to 200mg daily) and increase at weekly intervals by 100mg until a serum uric acid concentration of 360 $\mu\text{mol/L}$ (6mg/100ml) or less is attained but without exceeding the maximal recommended dose. Even with adequate therapy, it may require several months to deplete the uric acid pool sufficiently to achieve control of the acute episodes. Allopurinol is not effective for the treatment of acute gout attacks since it has no anti-inflammatory action and may intensify and prolong inflammation during the acute phase. In condition where the body's miscible urate pool is greatly increased (e.g. malignant disease and its treatment; Lesch-Nyhan syndrome), the reduction of urate formation by allopurinol is accompanied by a relative rise in the xanthine and hypoxanthine fractions. In these circumstances, the absolute concentration of xanthine could rise to a level at which deposition in the urinary tract may occur. The risk may be minimized by adequate hydration to achieve maximum diuresis. Alkalinization, of considerable benefit in relation to urate stones may be less so in relation to xanthine stones.

Drug Interactions: Amoxycillin or Ampicillin: Concurrent ampicillin or amoxycillin and allopurinol therapy has resulted in an increased incidence of drug induced skin rash. It is not clear whether this is due to allopurinol therapy. Azathioprine or Mercaptopurine: See Warnings. Chlorpropamide: In the presence of allopurinol, there may be competition in the renal tubule for the excretion of chlorpropamide. When renal function is poor, the recognized risk of chlorpropamide's prolonged hypoglycemic activity may be increased if allopurinol is given concomitantly.

Coumarin Anticoagulants: Under experimental conditions, allopurinol prolongs dicumarol's half-life. The clinical significance of this has not been established but this interaction should be kept in mind when allopurinol is given to patients already on anticoagulant therapy and the coagulation time should be reassessed.

Cyclophosphamide: Concurrent cyclophosphamide and allopurinol therapy may increase the incidence of bone marrow depression as compared with cyclophosphamide but the mechanism for this interaction is not known.

Theophylline: Large daily dose (600mg/day) of allopurinol may decrease theophylline clearance when both drugs are used for longer than 2 weeks. Since increases in serum theophylline concentrations of 25% have been reported, some patients may require monitoring for signs of possible theophylline toxicity and may need lower doses during concurrent allopurinol therapy. Thiazides: Hypersensitivity has occasionally been reported in

patients with renal compromise receiving allopurinol and thiazides concurrently. Administer this combination with caution. **Uricosurics:** Concomitant administration of a uricosuric agent with allopurinol may result in a decrease in urinary excretion of oxypurines and an increase in urate excretion compared to the excretion of these with allopurinol alone. Although the renal clearance of allopurinol is rapid, that of oxypurinol is slow and parallels that of uric acid, but is higher by a factor of about 2. The clearance of oxypurinol is increased by uricosuric drugs and as a consequence the addition of a uricosuric agent may reduce the extent of inhibition of xanthine oxidase by oxypurinol. However, such combined therapy may be useful in achieving minimum serum uric acid concentrations provided that total urinary uric acid load does not exceed the competence of the patients' renal function.

Adverse Effects: **Dermatological:** Skin rash, usually maculopapular is the most commonly reported adverse effect. Incidence of skin rash may be increased in the presence of renal disorders. In some instances, rashes have been followed by severe hypersensitivity reactions. Withdraw allopurinol immediately if such reactions occur (see Warnings). Exfoliative, urticarial or purpuric lesions, Steven-Johnson's syndrome (erythema multiforme) bullae and toxic epidermal necrolysis have also been reported. A few cases of alopecia with or without accompanying dermatitis have been reported. After recovery from mild reactions allopurinol may, if desired, be reintroduced at a small dose (e.g. 50mg/day) and gradually increased. If the rash recurs, withdraw allopurinol permanently. The drug should not be reinstituted in patients who have had a severe reaction. **Generalized Hypersensitivity:** Skin reactions associated with exfoliation, fever, chills, nausea and vomiting, lymphadenopathy, arthralgia and/or eosinophilia have occurred. If they do occur, it may be at any time during treatment and allopurinol should be withdrawn immediately and permanently. Corticosteroids may be beneficial in overcoming such reactions. When generalized hypersensitivity reactions have occurred, renal and/or hepatic disorders have usually been present. Hypersensitivity has been reported in patients with compromised renal function receiving allopurinol and thiazides concomitantly. **Gastrointestinal:** Diarrhea, intermittent abdominal pain, nausea and vomiting were reported. Gastrointestinal disorders diminish if allopurinol is taken after meals. **Blood and Lymphatic:** There have been occasional reports of reduction in the number of circulating formed elements of the blood including bone marrow suppression, granulocytopenia and thrombocytopenia usually in association with renal and/or hepatic disorders or in whom concomitant drugs have been administered which have a potential causing these reactions. **Miscellaneous:** The following adverse effects have been reported occasionally: fever, general malaise, headache, vertigo, somnolence, taste perversion, hepatic necrosis, granulomatous hepatitis, abnormal liver function tests, rise in BUN hyperlipemia, visual disorder, cataracts, macular changes, neuropathy, impotence, diabetes mellitus, furunculosis, hypertension, hematuria, edema, drowsiness, peripheral neuritis. Cataracts have been observed in some patients who have received allopurinol.

Overdose: **Symptoms:** Nausea and vomiting. **Treatment:** No treatment is normally required provided allopurinol is withdrawn and adequate hydration is maintained to facilitate the drug's excretion. If however, other forms of acute distress are observed gastric lavage should be considered otherwise the treatment is symptomatic.

Dosage:

Adults: Dosage range of 100 to 800mg daily divided into 1 to 3 doses. Single dose should not exceed 300mg. Allopurinol is better tolerated when taken with meals. In all patients receiving allopurinol, a high fluid intake (e.g. 2.5 to 3 litres) and the maintenance of a neutral or preferably slightly alkaline urine are recommended.

Gout: The dose of allopurinol varies with the severity of the disease and should be adjusted according to the response and tolerance of the patient. Some investigators have reported an increase in acute attacks of gout during the early stages of allopurinol administration. Accordingly, the patient should start with a low dose of allopurinol (100 to 200mg/day) and increase at weekly intervals by 100mg until a serum uric acid concentration of about 360µmol/L (6mg%) or less is attained or until the maximum recommended dosage of 800mg/day (in patients with normal renal function) is reached.

Also, a maintenance dose of colchicine (0.6mg twice daily) or a nonsteroidal anti-inflammatory drug should be given prophylactically when allopurinol is begun. After serum urate concentrations are controlled, it may be possible to reduce dosage; the minimum effective dose is 100 to 200mg/day. The average maintenance dosage is 200 to 300mg/day for patients with mild gout, 400 to 600mg/day for patients with moderately severe tophaceous gout and 700 to 800mg/day in severe conditions.

Dosage in renal impairment: Since allopurinol and its metabolites are excreted only by the kidney, drug accumulation can occur in renal failure and the initial dose of allopurinol should consequently be reduced. With a creatinine clearance of 0.33 to 0.17ml/s (20 to 10ml/min), a daily dosage of 200mg of allopurinol is suitable. When the creatinine clearance is less than 0.17ml/s (10ml/min) the daily dosage should not exceed 100mg. With extreme renal impairment (creatinine clearance less than 0.05ml/s (3ml/min), the interval between doses may

also need to be lengthened. Some clinicians recommend the following maintenance dosages of allopurinol based on the patient's creatinine clearance (refer to the blue pages for estimation of the creatinine clearance):

<u>Creatinine (ml/sec)</u>	<u>Clearance (ml/min)</u>	<u>Maintenance Dosages</u>
0	0	100mg every 3 days
0.17	10	100mg every 2 days
0.33	20	100mg daily
0.67	40	150mg daily
1.00	60	200mg daily
1.33	80	250mg daily
1.67	100	300mg daily
2.00	120	350mg daily
2.33	140	400mg daily

As no simple method of measuring allopurinol's blood concentrations is available, the correct size and frequency of dosage for maintaining the serum uric acid just within the normal range is best determined by using the serum uric acid concentration as an index. Once the daily dose of allopurinol necessary to produce the desired serum uric acid concentration has been determined, continued this dose until the serum uric acid concentration indicates a need for dosage adjustment. Normal serum urate concentrations are achieved in 1 to 3 weeks. The upper limit of normal is about 360µmol/L (6mg%) for men and postmenopausal women and 300µmol/L (5mg%) for premenopausal women. By the selection of the appropriate dose together with the use of uricosuric agents in certain patients, it is possible to reduce the serum uric acid concentration to normal and if desired to hold it as low as 120 to 180µmol/L (2 to 3mg%). Combined therapy of allopurinol and uricosurics will often result in a dosage reduction of both agents. In patients who are being treated with uricosuric agents, colchicine and/or anti-inflammatory agents, it is wise to continue this therapy while adjusting the allopurinol dosage until a normal serum uric acid concentration and freedom from acute attacks have been maintained for several months. If desired, the patient may then be transferred to allopurinol therapy exclusively. When an uricosuric agent is being withdrawn, dosage of the uricosuric agent should be gradually reduced over several weeks.

Prevention of Uric Acid Nephropathy during the Vigorous Therapy of Neoplastic Disease: Treatment with 600 to 800mg daily for 2 or 3 days prior to chemotherapy or irradiation is advisable. When allopurinol is used with mercaptopurine or azathioprine, dosage of the latter drugs must be reduced. (See Warnings). Continue treatment at a dosage adjusted to the serum uric acid concentration until there is no longer a threat of hyperuricemia and hyperuricosuria. Allopurinol treatment can be maintained during the antimitotic therapy for prophylaxis of the hyperuricemia which may arise during the natural crises of the disease. In prolonged treatment, 300 to 400mg of allopurinol daily is usually enough to control the serum uric acid concentration.

Prophylaxis of Renal Calcium Lithiasis: 200 to 300mg daily. Therapy should be continued indefinitely. Some patients have received maintenance doses of 200 to 300mg daily for more than 7 years. In some patients, the maintenance dose may be reduced to 100 to 200mg daily.

Children: For the treatment of secondary hyperuricemia associated with malignancies, (6 to 10 years of age): give 300mg of allopurinol daily; (<6 years of age): give 150mg of allopurinol daily. In the Lesch-Nyan syndrome, (6 to 10 years age): give allopurinol in doses of 10mg/kg/day. Evaluate the response after approximately 48 hours by monitoring serum uric acid and/or urinary uric acid concentrations and adjust the dose if necessary.

Supplied:

Apo-Allopurinol Tablets 100mg: White, round, biconvex, flat-edged tablet. Scored & engraved ALL over 100 on one side, other side plain. Available in PVC Blister of 100's and 1000's.

Apo-Allopurinol Tablets 300mg: Orange, round, biconvex, flat-edged tablet. Scored & engraved ALL over 300 on one side, other side plain. Available in PVC Blister of 100's and 500's.

Storage Conditions: Store below 30°C.

Manufacturer: Apotex Inc, 150 Signet Drive, Weston (Toronto), Ontario, Canada, M9L 1T9.

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